

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.2: Growth and Development in Plants — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 2.2: Growth and Development in Plants
Driving Question	DRIVING QUESTION: What causes the differences in germination, emergence and growth of maize and bean plants?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Farmer's Puzzle: Why Do Maize and Beans Behave Differently After the Same Rain?	In the time-lapse videos and pictures of maize and beans planted together after rain, we observed that the seeds germinated at different times, the root and shoot emerged, stems became taller and thicker, roots went deeper, some plants developed branches, and some seeds did not germinate even though they looked healthy.	We learned that the same rain and same planting day do not give the same result for every seed, and that we do not yet know what causes these differences. We drew our first models and learned that ideas can change when we find evidence. We also learned that we need to understand what growth and development mean.	Our first explanation (not yet marked): maize and bean plants may grow differently because of differences in their seeds and in what each seed needs, such as water, air and warmth. We do not yet have evidence, so we have posted our questions on the Driving Question Board. Our next question is: What is growth and development?
Lesson 2 Meaning of Growth and Development; Structure of Monocot and Dicot Seeds	We observed and dissected soaked maize grains and bean seeds, and saw that maize has one cotyledon (scutellum) with a large endosperm and protective coleoptile and coleorhiza, while beans have two large cotyledons, a testa, hilum, micropyle, plumule and radicle.	Growth is a permanent increase in size, dry mass or cell number, while development is the change in form and function as cells differentiate. Seed structures have specific functions such as protection, food storage, absorbing water and forming the new root and shoot.	Maize and beans are built differently, so they store and supply food to the embryo in different ways, which may partly explain differences in emergence and early growth. Since some healthy-looking seeds still failed to germinate, other factors, such as seed dormancy, must also be investigated.
Lesson 3 Seed Dormancy: Why Do Some Healthy-Looking Seeds Refuse to Germinate?	We examined maize, bean and other local seeds with hand lenses, noting hard and thin seed coats and how much water the seeds absorbed. We set up control, scarified, warm-water-soaked and leached/chilled seeds and recorded changes over time.	Seed dormancy is a resting state in which a living seed fails to germinate even when conditions seem suitable. Causes include a hard impermeable seed coat, chemical inhibitors, an immature embryo, and unsuitable environmental cues. Dormancy can be broken by scarification, soaking in warm water, leaching, chilling (stratification) and after-ripening. Dormancy helps seeds	Some seeds in the farmer's field do not germinate or are delayed even after the same rain because they are dormant: their hard seed coats may block water entry, inhibitors may need to be washed away or broken down, or their embryos may not be ready. Evidence: in our containers, treated seeds took up water and responded sooner than untreated controls. This explains part

		survive unfavourable seasons and spreads germination over time.	of the difference in germination between maize and beans, but we still need to find out how germination occurs once dormancy is broken.
Lesson 4 Germination: Process, Types and Conditions Necessary for Germination in Maize and Bean Seeds	We set up maize and bean seeds in containers with different conditions (moist and open to air, dry, submerged with little oxygen, cool, and boiled seeds) and compared them over several days. We also examined and drew germinating maize and bean seedlings. Record the number of seeds germinated in each container and the differences between how the maize and bean seedlings emerged.	Seeds need water, oxygen and a suitable temperature to germinate, and they must be alive (viable) and not dormant. Germination starts with water uptake, then the radicle emerges, then the plumule. Maize shows hypogeal germination (cotyledon stays below ground), while bean shows epigeal germination (cotyledons are lifted above ground).	Seeds planted on the same day after the same rain can germinate at different times, or fail, because of differences in moisture, oxygen (for example in waterlogged soil), temperature, seed viability and dormancy. Maize and beans also emerge differently because of their type of germination. After emergence, how do plants show growth? This will be investigated in the next lesson.
Lesson 5 Primary and Secondary Growth: How Do Plants Show Growth?	We observed prepared slides, photomicrographs or hand-cut sections of young bean (dicot) and maize (monocot) stems and a cut log from a woody tree with annual rings. We saw vascular bundles in a ring in the bean stem with cambium between xylem and phloem, scattered bundles in maize with no cambium, and secondary xylem, bark and rings in the older woody stem.	We learned that primary growth is an increase in length produced by apical meristems at shoot and root tips, while secondary growth is an increase in girth produced by the vascular cambium and cork cambium in dicots and woody plants. Maize (a monocot) mainly shows primary growth, whereas bean and trees such as Grevillea and Eucalyptus show both. This raises the question: How can you measure growth in plants?	Using Claim-Evidence-Reasoning, we explained that part of the difference in how maize and bean plants grow after germination is due to stem structure: the bean stem has cambium that allows thickening and branching, while the maize stem lacks cambium and grows mainly in length. This adds to our model that differences in germination and growth depend on seed structure, germination conditions, and the meristems that drive growth.
Lesson 6 Where Does a Maize or Bean Plant Grow? Measuring Growth and Regions of Growth in Seedlings	We marked maize and bean seedlings at equal intervals and measured how the marks moved apart over time. We also measured seedling height at intervals and plotted a growth curve. The marks near the root tip and shoot tip moved apart the most, and the curve showed a slow start, a rapid middle phase, and a levelling-off phase.	Growth in length is concentrated in specific regions (apical meristems at the root and shoot tips, followed by elongation), while stems thicken through secondary growth in some plants such as beans. Growth can be measured and shown on a growth curve that is S-shaped. Maize and bean seedlings grow at different rates and in different ways.	The maize and bean plants planted on the same day differ in height, root depth and branching because growth occurs in particular regions and at different rates in each plant type and over time. The sigmoid growth curve and the types of growth (primary and secondary) help explain these differences. What controls the cells at the tips to divide and elongate is still unknown, which raises the next question: Do we have hormones in plants?
Lesson 7 Plant Hormones: What Controls Growth in Maize and Bean Seedlings?	We watched a video and read about the major plant hormones (auxins, gibberellins, cytokinins, abscisic acid and ethylene) and recorded where each is made and what it does in the plant.	Hormones are chemical messengers that coordinate germination, dormancy, shoot and root growth, branching and fruit ripening. They work together with environmental factors such as water, oxygen and temperature.	Differences in germination, emergence and growth between maize and bean plants can partly be explained by hormone balance: for example, gibberellins and abscisic acid affect dormancy and germination time, auxin affects shoot elongation and apical dominance, and cytokinins affect branching.

			To compare growth we now need to measure it.
Lesson 8 Measuring Growth: Seedling Height, Stem Width and Growth Curves of Maize and Beans	We measured the height and stem width of maize and bean seedlings over time and plotted growth curves. Both curves were S-shaped, with a slow start, a rapid growth phase and a levelling-off phase, but maize grew faster and taller than beans in the same period.	Growth can be measured as height, stem width, mass or leaf area. A growth curve shows that growth rate changes over time (lag, exponential and stationary phases). Maize and beans differ in growth rate and pattern even when planted on the same day.	The growth rate changes because of food stores, cell division and elongation, leaf area for photosynthesis and the type of plant. Something inside the plant may be controlling the timing and speed of growth, so we ask: Do we have hormones in plants?
Lesson 9 Role of Hormones in Plant Growth and Development: Why Do Maize and Bean Plants Grow Differently?	We observed that a bean seedling with its tip removed grew more side branches but was shorter, while the intact seedling grew taller with few branches. We also saw (in the video or chart) how plant parts respond to hormones, e.g. bending toward light and rooting of cuttings, and we read that gibberellins, abscisic acid, cytokinins, auxins and ethylene each control different parts of growth.	We learned that hormones are chemical messengers that control germination, dormancy, cell elongation, branching, rooting and fruit ripening. Auxin from the shoot tip causes apical dominance, and removing the tip releases side buds. Farmers use this knowledge when pruning tea in Kericho, rooting cuttings in tree nurseries and ripening bananas and mangoes.	Hormones help explain why maize and beans planted together on the same day can germinate, emerge and grow differently: the balance of gibberellins and abscisic acid affects whether a seed breaks dormancy, and auxin and cytokinins control shoot growth and branching, with differences in species and seed condition changing the outcome. These internal signals work together with water, oxygen, temperature and seed structure to produce the differences we saw in the farmer's field.
Lesson 10 Synthesis and Application: Explaining the Farmer's Maize and Beans and Planning the Tree Nursery	We observed that maize and bean seeds differ in structure (monocot with endosperm vs dicot with cotyledons), that seeds failed to germinate without water, oxygen or warmth or when dormant, that maize germinates hypogeally and bean epigeally, that seedling height follows an S-shaped growth curve, and that stem cuttings treated with natural hormone sources such as bulb onion, Aloe or cinnamon rooted differently from the control.	We learned that germination, emergence and growth depend on seed structure and food stores, dormancy and its breaking, water, oxygen and suitable temperature, type of germination, growth at meristems (primary and secondary growth), and plant hormones such as auxins, gibberellins, cytokinins and abscisic acid, including apical dominance. We also learned that growth is an increase in size while development involves changes in form and function.	The farmer's maize and beans differed because each seed had a different structure, dormancy status and germination type, and experienced different water, oxygen and warmth conditions around the seed. These factors changed the levels and action of hormones such as gibberellins and abscisic acid in the seed and auxins in the growing plant, which control when the embryo grows, how the seedling emerges, and how the stem and roots grow. We can apply this to grow tree seedlings from cuttings by using natural rooting hormone sources, and share the seedlings to benefit our community and help reduce global warming.

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